AN EQUIVALENCE OF ABELIAN CATEGORIES INDUCES AN EQUIVALENCE OF BOUNDED BELOW DERIVED CATEGORIES OF WEAK SERRE SUBCATEGORIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the passage to derived categories and the verification of the quasi-inverse relation are both inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Weak Serre subcategory of a category of modules and the derived category of such a subcategory. Same chapter, the definition of a weak Serre subcategory and the bounded below derived category whose objects are complexes with cohomology in the subcategory. Supplies the two derived categories compared in the statement.
- Flatness of the morphism and the adjunction between pullback and push-forward. Same chapter, the adjunction f^* – Rf_* and the exactness of f^* under the flatness hypothesis. Supplies the functors and the compatibility used to compare the two derived categories.

Source reference. Original lemma 0D7T: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6368–6368. Navigation pointer invoked at proof line 6412; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0D7Q: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6286–6286. Navigation pointer invoked at proof

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line 6387; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0D7S: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6344–6344. Navigation pointer invoked at proof line 6426; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0D7R: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6315–6315. Navigation pointer invoked at proof line 6364; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 07RB: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6340–6340. Navigation pointer invoked at proof line 6340; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : (Sh(\mathcal{C}), \mathcal{O}) \to (Sh(\mathcal{C}'), \mathcal{O}')$ be a morphism of ringed topoi. Let $\mathcal{A} \subset Mod(\mathcal{O})$ and $\mathcal{A}' \subset Mod(\mathcal{O}')$ be weak Serre subcategories. Assume*

(1) *f is flat,*
(2) *f^* induces an equivalence of categories $\mathcal{A}' \to \mathcal{A}$,*
(3) *$\mathcal{F}' \to Rf_*f^*\mathcal{F}'$ is an isomorphism for $\mathcal{F}' \in \mathrm{Ob}(\mathcal{A}')$.*

Then $f^* : D^+_{\mathcal{A}'}(\mathcal{O}') \to D^+_{\mathcal{A}}(\mathcal{O})$ is an equivalence of categories with quasi-inverse given by $Rf_* : D^+_{\mathcal{A}}(\mathcal{O}) \to D^+_{\mathcal{A}'}(\mathcal{O}')$.

Proof. By assumptions (2) and (3) and Lemmas 0D7T and 0D7Q we see that $f^* : D^+_{\mathcal{A}'}(\mathcal{O}') \to D^+_{\mathcal{A}}(\mathcal{O})$ is fully faithful. Let $\mathcal{F} \in \mathrm{Ob}(\mathcal{A})$. Then we can write $\mathcal{F} = f^*\mathcal{F}'$. Then $Rf_*\mathcal{F} = Rf_*f^*\mathcal{F}' = \mathcal{F}'$. In particular, we have $R^pf_*\mathcal{F} = 0$ for $p > 0$ and $f_*\mathcal{F} \in \mathrm{Ob}(\mathcal{A}')$. Thus for any $K \in D^+_{\mathcal{A}}(\mathcal{O})$ we see, using the spectral sequence $E_2^{p,q} = R^pf_*H^q(K)$ converging to $R^{p+q}f_*K$, that Rf_*K is in $D^+_{\mathcal{A}'}(\mathcal{O}')$. Of course, it also follows from Lemmas 0D7S and 0D7R that $Rf_* : D^+_{\mathcal{A}}(\mathcal{O}) \to D^+_{\mathcal{A}'}(\mathcal{O}')$ is fully faithful. Since f^* and Rf_* are adjoint we then get the result of the lemma, for example by Categories, Lemma 07RB. □